

ELECTRICAL CIRCUIT AND NETWORKS

Diploma in Electrical Engineering

Semester - III

SBTE, Bihar

J) Theory Session Outcomes (TSOs) and Units: T2420301

Major Theory Session Outcomes (TSOs)	Units	Relevant COs Number(s)
TSO 1a. Classify the given components into active and passive. TSO 1b. Differentiate between the following: - Linear and nonlinear, - Unilateral and bilateral, - Lumped and distributed parameters, - Independent and dependent source TSO 1c. Describe the nature of the voltage-current characteristics of the given type of voltage source TSO 1d. Reduce the given passive network by using star-delta transformation TSO 1e. Determine the equivalent resistance by using star/delta transformation from voltage to current transformation and vice versa. TSO 1f. Apply Kirchhoff's Current and Voltage Law to analyze the given electric circuit(s). TSO 1g. Apply mesh analysis and nodal analysis to determine the current and voltage in a given circuit. TSO 1h. Explain Transient and steady state condition in a given circuit TSO 1i. Determine initial state and steady state of R, L & C in DC Circuit and give its application	Unit-1.0 Basics of Electrical Circuits 1.1 Definition of: - Linear and non-linear - Active and passive components - Unilateral and bilateral - Lumped and distributed parameters - Independent and dependent source 1.2 Concept of open circuit, closed circuit and short circuit, node, branch, mesh and loop 1.3 Star/Delta transformation of passive network 1.4 Source transformation 1.5 Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL), Applications 1.6 Mesh Analysis and Nodal analysis of networks 1.7 Transient & steady state 1.8 Concept of initial and final conditions in switching circuits, Meaning of $t=0^-$, $t=0^+$, and $t = \infty$. R, L and C at initial and final conditions	CO1
TSO 2a. Solve a given complex linear active bilateral electrical circuit/network with multiple source using superposition theorem. TSO 2b. Substitute a given complex electrical circuit/network across its load terminals by a equivalent circuit comprising of a voltage source in series with Thevenin's resistance. TSO 2c. Simplify a given linear complex electrical circuit/network to an equivalent circuit comprising of current source in parallel with resistance. TSO 2d. Transfer supply voltage and output current mutually in a linear passive electrical circuit/network by applying reciprocity theorem and also mention its applications TSO 2e. Determine the conditions for maximum power transfer across Load	Unit-2.0 Network Theorems: 2.1 Superposition theorem and its applications 2.2 Thevenin's theorem and its applications 2.3 Norton's theorem and its applications 2.4 Reciprocity theorem and its applications 2.5 Maximum power transfer theorem 2.6 Application of theorems to solve DC networks	CO1, CO2
TSO 3a. Explain the behavior of AC voltage, current and power through pure resistance, pure inductance and pure capacitance with sketches	Unit-3.0 Single Phase AC Circuits: 3.1 Generation of an alternating EMF	CO3

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TSO 3b. Determine the current and voltage, impedance of the given series/parallel RL/RC/LC/ RLC circuit TSO 3c. Determine the active, reactive, apparent power and power factor of the given AC circuit Differentiate the given AC circuit quantities. TSO 3d. Represent the given AC circuit quantities in complex form. TSO 3e. Convert the given AC quantity in rectangular to polar and vice versa and other arithmetic operations. TSO 3f. Determine the current and voltage, impedance of the given series/parallel RL/RC/ RLC circuit TSO 3g. Determine the active, reactive, apparent power and power factor of the given AC circuit	3.2 AC circuit quantities: Peak value, RMS and Average value of a Sinusoidal voltage waveform 3.3 Vector representation of an alternating quantity, addition, subtraction, multiplication and division, Conversion from rectangular to polar and vice versa and exponential form 3.4 Waveforms, phasor diagram and expression of voltage, current and power in pure: Resistance, Inductance, Capacitance 3.5 AC Series and parallel circuits, Phasor diagrams and impedance triangle 3.6 Active, reactive, apparent power with examples, Power factor, lagging, leading and unity power factor, effects of poor power factor, power triangle	
TSO 4a. Explain the phenomena of resonance in the given RLC series and parallel circuit with sketches TSO 4b. Determine the resonant frequency of the given series RLC circuit TSO 4c. Explain the significance of quality factor of the given series RLC circuit. TSO 4d. Determine the equivalent impedance and current magnitude of the given parallel RLC circuit under resonance condition TSO 4e. Explain 2-port network and classify it. TSO 4f. Determine the Z, Y & T parameter using electrical circuits TSO 4g. Determine the conditions of symmetry and reciprocity of two port networks	Unit-4.0 Resonance and Two Port Network: 4.1 Resonance and its importance in electrical circuit 4.2 Series & Parallel resonance: Resonant frequency, Quality factor, bandwidth and selectivity in series & Parallel RLC circuit. 4.3 Magnification in series and parallel resonant circuit 4.4 Comparison of series and parallel resonance and its applications 4.5 Significance of two port network and its types 4.6 Open Circuit (Z) parameter, Short Circuit (Y) parameter & Transmission parameter 4.7 T and Pie representation of circuits Symmetry and reciprocity of two port network	CO3, CO4
TSO 5a. Explain phase and time displacement of three phases. TSO 5.b Determine the current drawn by the given three phase balanced load connected in star/delta. TSO 5.c Explain the line and phase relation of current /voltage of three phase circuit. TSO 5.d Determine the current drawn by the given three phase loads connected in parallel. TSO 5.e Determine the power and power factor of the given three phase load using two wattmeters TSO 5.f Determine the power factor of the given type of three phase load connected in parallel using power triangle.	Unit-5.0 Three phase AC circuits 5.1 Generation of three phase voltage 5.2 Three phase three wire source and three phase four wire source, Phase sequence and phasor diagram 5.3 Line and phase relationship: Star/Delta 5.4 Three phase load -balanced and unbalanced Load 5.5 Measurement of power in three phase circuits	CO4, CO5

ELECTRICAL MEASUREMENTS AND INSTRUMENTATION

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J) Theory Session Outcomes (TSOs) and Units: T2420302

Major Theory Session Outcomes (TSOs)		Units	Relevant COs Number(s)
TSO.1.a	Explain basic concept and significance of measurements.	Unit-1.0 Basics of Measurement and Instrumentation - Measuring systems and requirements: Block diagram - Deflecting, controlling and damping torque - Accuracy, precision, Resolution, tolerance, sensitivity and repeatability - Errors in measurement - Types of errors- Limiting error, Gross error, systematic Error, Random Error, Guaranteed accuracy error - Classification of basic instruments -Indicating, Recording and Integrating type	CO1
TSO.1.b	Differentiate between deflecting, controlling and damping torque in an instrument.		
TSO.1.c	Explain the given terms related to measuring systems.		
TSO.1.d	Classify types of errors in measurement.		
TSO.1.e	Classify different types of instruments.		
TSO.2.a	Explain the general principle of measuring current, voltage in an electrical system.	Unit-2.0 Measurement of Voltage and Current - Current and voltage measurement: Principle - Galvanometer, Ammeter, Voltmeter - Calibration of ammeter and voltmeter - Range Extension of ammeter and voltmeter using: - Shunts and Multipliers - Current Transformer (CT) and Potential Transformer (PT) (Construction, working and applications) - Permanent Magnet Moving Coil (PMMC), Moving Iron, Induction, Dynamometers type instruments: Working principle, construction, applications, merits and demerits - Essentials and advantages of electronic instruments - True RMS reading voltmeter. - Digital Voltmeters (DVM) and its types - Digital multimeters- Block diagram	CO2
TSO.2.b	Extend the range of ammeter and voltmeter using shunt, multipliers and Current Transformer and Potential Transformer		
TSO.2.c	Describe the calibration procedure of the given meters.		
TSO.2.d	Describe the construction and working principle of the given types of electromechanical measuring instruments.		
TSO.2.e	Explain the working principle of the given types of electromechanical measuring instruments.		
TSO.2.f	Explain the significance of using electronic instruments		
TSO.2.g	Explain the working of true rms voltmeter with block diagram.		
TSO.2.h	Explain the working of DVM with block diagram.		
TSO.2.i	Describe working and advantage of digital multi meter with block diagram.		
TSO.3.a	Explain the general principle of measuring power and energy in an electrical system	Unit-3.0 -Measurement of Power and Energy - Power and energy Measurement: Principle - Measurement of single and three phase power using wattmeter - one wattmeter and two wattmeter method - Effect of power factor variation on wattmeter reading in two wattmeter method - Measurement of energy using single phase and three phase watt-hour/Energy meter - Errors and compensation in energy meter - Cripping and phantom loading in energy meter - Calibration of watt meters and energy meters - Digital energy meter: Block diagram, Working	CO3
TSO.3.b	Explain the working of induction and dynamometer type instruments.		
TSO.3.c	Explain the general principle of measuring single and three phase power in an electrical system		
TSO.3.d	Apply the power measuring technique in distribution and transmission system.		
TSO.3.e	Describe the construction and working of single and 3 phase energy meters		
TSO.3.f	Determine errors and compensation in an energy meter.		

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Major Theory Session Outcomes (TSOs)		Units	Relevant COs Number(s)
TSO.3.g	Describe the standard procedure for calibration of the given equipment.		
TSO.4.a	Classify the resistance into low, medium and high.	Unit 4.0-Measurement using Bridges/Meters - Classification of resistances - Low, Medium and High - Concept of bridge, balancing - Resistance measurement -Kelvins double bridge (Low), voltmeter - ammeter method, Wheatstone bridge, ohmmeter (Medium) and Megger (High). - Earth resistance - Inductance measurement – Anderson, Maxwell inductance capacitance bridge - Capacitance Measurement- Schering bridge. - Frequency Measurement -Wein Bridge, Weston Frequency meter.	CO4
TSO.4.b	Explain the basic concept of bridge and bridge balancing.		
TSO.4.c	Describe the procedure to measure unknown resistance (low, medium and high) using appropriate bridge/method.		
TSO.4.d	Describe the method of using Earth tester to measure earth resistance		
TSO.4.e	Describe the procedure to measure an unknown inductance by using appropriate bridge		
TSO.4.f	Describe the procedure to measure the unknown Capacitance using appropriate bridge		
TSO.4.g	Describe the procedure to measure an unknown frequency by using Wein Bridge/ Weston Frequency meter.		
TSO.5.a	Describe the procedure to measure the power factor by using P.F meter.	Unit 5.0- Measurement of Other Electrical Parameters - Dynamometer, Power Factor meter - Phase sequence indicator: rotating type - Synchro scope - CRO-basic clock diagram, Cathode Ray Tube, Electrostatic and magnetic deflection, X & Y Amplifiers, Controls on CRO and their functions, Lissajous pattern - Measurement of voltage, amplitude, time period, frequency and phase angle - Digital Storage Oscilloscope- Basic block diagram and working - Q/LCR meter, Digital LCR meter- Block diagram, Working principle - Analog/Digital recorders, Graphic recorder, Strip Chart recorder, XY recorder (Only block diagram) - Applications of Recorders.	CO5
TSO.5.b	Describe the procedure of using a Phase sequence indicator to determine the phase sequence		
TSO.5.c	Explain the use of Synchroscope in synchronization		
TSO.5.d	Describe functions of basic building blocks of CRO		
TSO.5.e	Explain deflection systems of CRO		
TSO.5.f	Explain working of digital storage oscilloscope using block diagram.		
TSO.5.g	Describe the working of Q/ LCR meter using block diagram		
TSO.5.h	Explain the working of the given type of recorders with the help of a block diagram		



DC MACHINES AND TRANSFORMERS

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J) Theory Session Outcomes (TSOs) and Units: T2420303

Major Theory Session Outcomes (TSOs)	Units	Relevant COs Number(s)
TSO.1.a Describe the construction details of a DC machine. TSO.1.b Explain the working principle of the DC Generator. TSO.1.c Classify DC Generator. TSO.1.d Explain effect of armature reaction in a DC machine. TSO.1.e Explain the process of commutation in DC Generator. TSO.1.f Describe the procedural steps of voltage build up in the DC generator. TSO.1.g Interpret the internal and external Characteristics of the given DC Generators.	Unit-1.0 DC Generators 1.1 DC Machine: Construction 1.2 DC Generator: Working Principle 1.3 Types of DC generators: Shunt, Series and Compound and their applications 1.4 EMF equation 1.5 Armature reaction and its effects. 1.6 Commutation. 1.7 Voltage build-up in DC Generators 1.8 Internal and External Characteristics. 1.9 Losses and Efficiency	CO1
TSO.2.a Explain the working principle of DC motor. TSO.2.b Determine the torque for a given DC motor TSO.2.c Interpret the various characteristics of DC motor TSO.2.d Describe the procedure to start the DC motor using given type of starters. TSO.2.e Describe the procedure to control the speed of the given DC motor. TSO.2.f Describe the procedure to select a particular DC motor for the given application. TSO.2.g Enlist the applications of different types of motors including applications in modern transportation systems.	Unit-2.0 DC Motors 2.1 Working Principle. 2.2 EMF equation, significance of Back EMF 2.3 Torque and Speed 2.4 Characteristics of DC motors - Torque vs Speed, Flux vs Current - Torque vs current, Speed vs Current 2.5 Need of Starters -3-point and 4-point starters. 2.6 Losses and Efficiency 2.7 Speed Control of DC motors - Armature Control Method - Field Flux Control Method 2.8 Selection of DC Motors. 2.9 Applications of different types of DC motors	CO1, CO2
TSO.3.a Describe the constructional details of a single-phase transformer. TSO.3.b Explain the working principle of transformer. TSO.3.c Derive EMF equation of a single-phase transformer. TSO.3.d Calculate various losses using Open circuit and Short circuit test. TSO.3.e Differentiate Ideal and Practical transformer. TSO.3.f Explain the performance with phasor diagram of the given transformer under no load and different loading conditions for unity, lagging and leading power factor TSO.3.g Explain the importance of voltage regulation. TSO.3.h Explain the condition of maximum efficiency of a transformer. TSO.3.i Describe the procedure of performing the Polarity test, Open circuit, Short Circuit Test in the given transformer. TSO.3.j Enlist the essential and desirable conditions for parallel operation of two single phase transformers.	Unit 3.0 Single Phase Transformer 3.1 Construction. 3.2 Types -Shell type and core type. 3.3 Working Principle, Useful and leakage flux 3.4 EMF Equation, Transformation Ratio. 3.5 Losses – Core and copper 3.6 Ideal and Practical transformer. 3.7 Equivalent circuit and phasor diagram. 3.8 Per Unit (pu) system in transformers 3.9 Voltage Regulation -Condition for maximum, zero and minimum regulation. 3.10 Efficiency -Condition for Maximum Efficiency, All day efficiency and its significance 3.11 Polarity test, Open circuit, Short Circuit Test 3.12 Parallel operation of two single transformers. 3.13 Autotransformers -Construction, application, Inductive and conductive power transfer, Copper saving.	CO3

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Major Theory Session Outcomes (TSOs)	Units	Relevant COs Number(s)
TSO.3.k Differentiate between autotransformer and two winding transformers. TSO.3.l Explain the significance of Amorphous Metal Transformer (AMT).	3.14 Energy efficient Amorphous Metal Transformer (AMT).	
TSO.4.a Describe the constructional details of 3-phase transformer. TSO.4.b Classify the different phasor groups of 3-phase transformer TSO.4.c Explain the working of open delta transformer. TSO.4.d Enlist the essential and desirable conditions for parallel operation of two 3-phase transformer. TSO.4.e Explain the Magnetization phenomenon in Transformer.	Unit 4.0 Three Phase Transformer 4.1 Bank of three single phase transformer and single unit of 3-phase transformer 4.2 Different parts of 3-phase transformers 4.3 Phasor groups of 3-phase transformers 4.4 Open Delta transformer 4.5 Parallel Operations of two 3-phase transformers 4.6 Magnetization phenomenon in Transformer.	CO3, CO4
TSO.5.a Explain the construction and working principle of the given special purpose transformers. TSO.5.b Describe the basic testing Procedure of the given special purpose transformers. TSO.5.c State the applications of the given special purpose transformers.	Unit 5.0 Special Purpose Transformers 5.1 Isolation Transformer 5.2 Grounding Transformer 5.3 Instrument transformer 5.4 Current Transformer 5.5 Potential Transformer 5.6 Welding transformer	CO4, CO5

Note: One major TSO may require more than one theory session/period.

ELECTRICAL POWER GENERATION TRANSMISSION AND DISTRIBUTION

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J) Theory Session Outcomes (TSOs) and Units: T2420304

Major Theory Session Outcomes (TSOs)	Units	Relevant COs Number(s)
TSO 1a. Describe the layout of the given type of power plant for electric power generation with labeled block diagram. TSO 1b. Explain the working of the given type of power plant with sketches. TSO 1c. Describe the criteria considered for site selection of the given power plant. TSO 1d. State the functions of different components of the given type power plant. TSO 1e. Explain the properties of the fuel used in the specified thermal power plants. TSO 1f. Describe the energy conversion process in the given type of power plants TSO 1g. State the given terms related to Power generation plants. TSO 1h. Explain the given factors which affects the economics of electric power generation.	Unit-1.0 Generation of Electrical Power 1.1 Thermal Power Plant: Lay out, working and Site selection. - Major Auxiliaries: Functions - Properties of conventional fuels used in Thermal Power Plants 1.2 Hydro Power Plants: Elements of Hydro power plant, Energy conversion process, Layout, and Site selection, Classification Nuclear power Plant: Lay out, working and site selection Properties of conventional fuels used in nuclear power plants, safe practices in nuclear power plants, Types of nuclear reactors, Disposal of nuclear waste and nuclear shielding. 1.4 Gas power plant: Lay out, working and site selection Other features of Power generation Plants: Connected load, firm power, cold reserve, hot reserve, spinning reserve. Factors affecting cost of generation: Average demand, maximum demand, demand factor, plant capacity factor, plant use factor, diversity factor, load factor and plant load factor. Base load and peak load plants; Load curve, load duration curve, integrated duration curve.	CO1
TSO 2a. Describe with sketches the layout and working of the solar energy power plant. TSO 2b. Describe with sketches the layout and working of the wind energy power plant. TSO 2c. Describe with sketches the layout and working of the Bio-mass based power plants. TSO 2d. Describe with sketches the layout and working of the Geo-thermal energy power plant. TSO 2e. Describe with sketches the layout and working of the Ocean energy power plant. TSO 2f. Describe the layout and working of the Fuel cell with sketches.	Unit-2.0 Renewable Energy Sources 2.1 Types of Renewable Energy Sources. 2.2 Solar Energy: Potential of solar energy, Photovoltaic effect, Construction & materials used in solar photo-voltaic cells, working & applications of solar energy. 2.3 Wind Energy: Selection of site for wind mills, Working Principle, Block diagram, Applications 2.4 Bio-Mass & Bio-Gas Energy: Composition of Bio-Gas & its calorific value, Traditional and non-traditional Biogas plants, Bio-mass based power generation plants, their capacities and applications 2.5 Geo-thermal Energy: Working Principle, Block Diagram and applications. 2.6 Ocean Energy: Ocean Thermal Electric Conversion, Energy from Tides, Site requirements, Advantages and Limitations of Tidal power generation. 2.7 Fuel Cells: Construction, working types and applications	CO1, CO2
TSO 3a. Explain the effects of R, L and C on the given transmission line. TSO 3b. Explain the features of given type of transmission lines TSO 3c. State the features of different types of conductors. TSO 3d. State the need for different types of insulators and calculate string efficiency	Unit-3.0 Transmission of Electrical Power 3.1 Transmission Line Parameters: Resistances, Inductances and Capacitances 3.2 Classification of Transmission Lines: short, medium and long 3.3 Comparison of different types of transmission systems.	CO2, CO3

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TSO 3e. of given type of insulators. Describe the given method of improving string efficiency of insulators. TSO 3f. Explain the criteria for spacing of conductors TSO 3g. Calculate sag in given type of transmission system. TSO 3h. Explain the phenomena of corona and factors affecting it in given type of transmission line. TSO 3i. Explain different phenomena occurred in given transmission line. TSO 3j. Calculate the performance parameters single phase short transmission line. TSO 3k. Explain the importance and functions of the load dispatch Centre. TSO 3l. State the salient features of the given EHV transmission system.	3.4 Types of Conductors-Copper, Aluminum: Solid, stranded and bundled conductors. 3.5 Line Insulators – requirements, types, Failure of insulator. 3.6 String Efficiency, string efficiency improvement method: By using longer cross arm, By grading the insulator and By using guard ring 3.7 Spacing between Conductors, span length and sag calculation. 3.8 Corona – corona formation, advantages & disadvantages, factors affecting corona 3.9 Skin effect, proximity effect, Ferranti effect and Transposition of conductors. 3.10 Losses, efficiency, regulation and Effect of load power factor 3.11 Load dispatch Centre 3.12 Issues of Distributed Generation Integrated to distribution Grid. 3.13 Requirement of EHV transmission. - HVAC Transmission - HVDC Transmission	
TSO 4a. State the need for distribution system. TSO 4b. Describe the various connection schemes of the distribution system with sketches. TSO 4c. Calculate voltage drop and minimum potential point using the given methods for 1-phase and 3-phase distribution system. TSO 4d. Describe the measures to be adapted to take of the distributed generation in the distribution system. TSO 4e. State the need for electrical substations and relevant site for given substation. TSO 4f. Sketch the elevation layout of a typical 11/33kV electrical substation with various switchgear and typical spacing between them and the ground level as well. TSO 4g. Describe the installation procedure of substation equipment's. TSO 4h. State the features of unarmored and armored underground cables TSO 4i. Describe the given laying procedure for underground cable.	Unit-4.0 Electrical Power Distribution System 4.1 Distribution system and its Requirements. 4.2 Connection schemes of distribution system. 4.3 A.C. distribution: Voltage drop, sending end voltage, receiving end voltage, point of minimum potential, minimum potential value and power loss 4.4 Power factor referred to receiving end voltage and power factor referred to respective load voltage. 4.5 Substation: Classification of substations based on; voltage level and Type of installation 4.6 Selection and location of substation 4.7 Installation of substation 4.8 Underground Cables: Reqs. classification, construction, comparison with overhead lines. 4.9 Laying of underground cable: Direct laying, Draw in system and Solid system	CO3, CO4
TSO 5a. State the concept and purpose of deregulation in power industry. TSO 5b. State the reasons for adopting deregulation in development of power industry. TSO 5c. Explain the process involved in restructured power system. TSO 5d. State the functions of the given ancillary support system. TSO 5e. State issues involved in deregulation process.	Unit-5.0 Deregulated Power system 5.1 Introduction. 5.2 Reasons for restructuring/ deregulation of power system. 5.3 Objectives of deregulation of various power system across the world. 5.4 Restructuring process 5.5 Ancillary services: Frequency support ancillary services, voltage control ancillary services and black start ancillary services 5.6 Issues involved in deregulation	CO4, CO5

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PYTHON PROGRAMMING

J) Theory Session Outcomes (TSOs) and Units: T2418305

Major Theory Session Outcomes (TSOs)	Units	Relevant COs Number(s)
TSO 1a. Differentiate between Procedure Oriented P and Object Oriented Programming approach with example. TSO 1b. Use the concept of Lvalue and Rvalue TSO 1c. Write python program using various data types and operators	Unit 1.0 Fundamentals of Python Programming Syntax 1.1 Introduction to Python Character Set, Python Tokens, Variables, Lvalue and Rvalue Concepts, and the Use of Comments. 1.2 Overview of Data Types: - Number Types: Integer, Floating Point, Complex - Boolean Type - Sequence Types: String, List, Tuple - None Type - Mapping Type: Dictionary - Distinction between Mutable and Immutable Data Types 1.3 Understanding Operators: - Arithmetic Operators - Relational Operators - Logical Operators - Assignment Operator - Augmented Assignment Operators - Expressions and Statements - Type Conversion and Input/Output Mechanisms - Precedence of Operators - Expression Evaluation	CO-1
TSO 2a. Write Python program using decision making statements TSO 2b. Write Python program using loop structure to solve iterative problems	Unit-2.0 Conditional and Iterative statements 2.1 Conditional statements: - simple if statement - if- else statement - if-elif-else statement 2.2 Iterative statements: - while loop - for loop - range function - break and continue statements - nested loops	CO-2
TSO 3a. Perform various operations on string using string operators and methods TSO 3b. Perform various operations on List using list operators and methods TSO 3c. Perform various operations on tuples using tuples operators and methods TSO 3d. Perform various operations on set using set methods	Unit-3.0 String, List, Tuples, set and Dictionary 3.1 String: - Indexing - string operations (concatenation, repetition, membership & slicing) - traversing a string using loops - Built-in functions. 3.2 Lists:	CO-3

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Major Theory Session Outcomes (TSOs)	Units	Relevant COs Number(s)
TSO 3e. Perform various operations on dictionary using dictionary methods	- Introduction - Indexing in list - list operations: concatenation, repetition, membership & slicing, traversing a list, built- in list functions, linear search on list of numbers and counting the frequency of elements in a list 3.3 Tuples: Creating, initializing, accessing elements, tuple assignment, performing operations on tuples, tuple methods and built-in functions, nested tuples 3.4 Set: Creating set, traversing, adding, removing data in set, performing set operations like join, Union intersection, difference 3.5 Dictionary: accessing items in a dictionary using keys, mutability of dictionary: adding a new item, modifying an existing item, built-in dictionary functions.	
TSO 4a. Create and use user defined functions to implement modular programming approach TSO 4b. Differentiate variable scope with example. TSO 4c. Import and use Python modules, libraries	Unit-4.0 Python Functions, Modules and packages 4.1 Functions: types of function (built- in functions, functions defined in module, user defined functions), creating user defined function, arguments and parameters, default parameters, positional parameters, Lambda functions, returning value, scope of a variable: global scope, local scope 4.2 Modules and Packages: Importing module using 'import' Regular Expressions, Exception Handling, PyPI Python Package Index, Pip Python package manager, Importing Libraries and Functions	CO-4
TSO 5a. Write simple Python programs using numpy TSO 5b. Use Numpy array in python program TSO 5c. Use Numpy to solve linear algebra problem.	Unit-5.0 Numpy 5.1 Introduction to NumPy 5.2 Installation of NumPy 5.3 NumPy Arrays: - Understanding the NumPy array - The fundamental data structure in NumPy. - Creation of arrays using different methods: np.array(), np.zeros(), np.ones(), etc. - Exploring array attributes like shape, size, and dimensions. 5.4 Array Indexing and Slicing: - Accessing elements and subarrays in NumPy arrays using indexing and slicing. - Demonstration of the difference between one-dimensional and multi-dimensional array indexing. 5.5 Array Operations:	CO-5

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	- Performing element-wise operations on NumPy arrays. - Exploring universal functions (ufuncs) for mathematical operations. 5.6 Linear Algebra with NumPy: - Introduction to linear algebra using NumPy. - Matrix multiplication, determinant, and solving linear equations. 5.7 File input and output with Numpy 5.8 Broadcasting in Numpy	
TSO 6a. Explain different types of Exceptions in python TSO 6b. Write Python programs for exception handling in Python TSO 6c. Differentiate different modes of file opening. TSO 6d. Perform read, Write, Append operations in files	Unit 6: Exception and File Handling in Python 6.1 Exception Handling: syntax errors, exceptions, need of exception handling, user-defined exceptions, raising exceptions, handling exceptions, catching exceptions, Try - except - else clause, Try - finally clause, recovering and continuing with finally, built-in exception classes. 6.2 File Handling: text file and binary file, file types, open and close files, reading and writing text files, reading and writing binary files, file access modes	CO-6

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